From fixed-energy decay to dynamical localization

A conditional weak-disorder argument for the square-lattice Anderson model

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Abstract

For the Anderson operator on $\mathbb{Z}^2$ with independent uniform potentials in $[-h, h]$, $0 < h \leq 1$, we prove a conditional transfer theorem: exponential fractional-moment decay on finite tori at each fixed interior energy implies exponential decay of expected eigenfunction correlators on the full spectrum, all polynomial dynamical moments, and an exponentially decaying eigenbasis. The proof has three steps: convert the torus estimate into a strict finite-volume test; add the two spectral endpoints and select a finite energy cover; apply spectral averaging and pass to the lattice. The quantitative torus estimate is assumed explicitly and attributed to OpenAI's 2026 preprint. Its constants may depend on energy. The proof does not require an energy-uniform input or deduce dynamical localization from pure-point spectral type alone.

1 Introduction: the result and its quantitative input

The main result is a precise implication from fixed-energy resolvent decay to localization of the full time evolution. We state it first, then distinguish the input taken from the cited preprint from the argument proved here. Throughout, the disorder is weak and fixed: $0 < h \leq 1$. There is no claim of uniform bounds as $h \downarrow 0$.

Let δ_x be the coordinate vector at $x \in \mathbb{Z}^2$, and let $(v_x)_{x\in\mathbb{Z}^2}$ be independent, each uniform on $[-h, h]$. Thus

$$\mathbb{P}_h = \bigotimes_{x\in\mathbb{Z}^2} \frac{\mathbf{1}_{[-h,h]}(v_x)}{2h}\, dv_x, \qquad (H_h u)(x) = \sum_{|y-x|_1=1} u(y) + v_x u(x). \tag{1.1}$$

This is a bounded self-adjoint operator on $\ell^2(\mathbb{Z}^2)$. Write $I_0 = [-4, 4]$ for the free band and $\Sigma_h = [-4 - h, 4 + h]$ for the deterministic spectral interval. The estimate $\|H_h\| \leq 4 + h$ gives $\mathbf{1}_{\Sigma_h}(H_h) = \mathrm{I}$, which is all that the argument needs. For a finite $\Omega \subset \mathbb{Z}^2$, the Dirichlet restriction deletes bonds leaving Ω; its Green function is the matrix entry

$$G_\Omega(x, y; z) = \langle \delta_x, (H_{h,\Omega} - z)^{-1} \delta_y \rangle. \tag{1.2}$$

We write $\mathbb{E}$ for expectation under the physical product law and any explicitly introduced independent auxiliary variables.

For a Borel set $I \subseteq \Sigma_h$, define

$$Q_I(x, y) = \sup_{\substack{f:\mathbb{R}\to\mathbb{C} \text{ Borel} \\ \|f\|_\infty \leq 1}} \left| \langle \delta_x, f(H_h)\mathbf{1}_I(H_h)\delta_y \rangle \right|. \tag{1.3}$$

The spectral theorem defines $f(H_h)$. Its cross spectral measure is $\mu_{xy}(B) = \langle \delta_x, \mathbf{1}_B(H_h)\delta_y \rangle$, and $Q_I(x, y) = |\mu_{xy}|(I) \leq 1$ is its total variation on I. Thus Q_I controls every bounded function of the operator at once, including $f(E) = e^{-itE}$ for all $t \in \mathbb{R}$. The desired quantitative conclusion is

$$\mathbb{E}Q_{\Sigma_h}(x, y) \leq C(h)e^{-\mu(h)|x-y|_1}, \qquad C(h) < \infty, \quad \mu(h) > 0. \tag{1.4}$$

Theorem 1.1 (Main result: dynamical localization from the torus estimate)**.** *Fix $h \in (0,1]$ and assume the quantitative input* (1.7) *at every $E \in (-4-h, 4+h)$, as stated in Hypothesis 1.2 below. Then* (1.4) *holds on the full spectrum. Put $q = 1/128$ and $p = q/2 = 1/256$. The finite scale $L = L_{\max}$ in Proposition 2.5 gives admissible constants*

$$\mu(h) = \frac{\log 2}{(2-p)(2L+2)}, \qquad C(h) = \max\left\{1, \left[\frac{32(4+h)\,2^{p-1}}{1-p} h^{-1}\right]^{1/(2-p)}\right\}. \tag{1.5}$$

For every $r \geq 0$ and $y \in \mathbb{Z}^2$,

$$\mathbb{E} \sup_{t\in\mathbb{R}} \sum_{x\in\mathbb{Z}^2} (1+|x-y|_1)^r |\langle \delta_x, e^{-itH_h}\delta_y\rangle|^2 < \infty. \tag{1.6}$$

Almost surely H_h has an orthonormal eigenbasis whose members decay exponentially in space.

Proof in three steps; details in Sections 2–3. First, Proposition 2.3 and Lemma 2.4 give a strict finite-volume test near every energy in Σ_h. Second, Proposition 2.5 selects finitely many tests; Theorem 2.2 then gives, uniformly for z near Σ_h,

$$\mathbb{E}|G_\Omega(x,y;z)|^p \leq 4K_p e^{-\mu_0 d_\Omega(x,y)}, \qquad \mu_0 = \frac{\log 2}{2L+2}.$$

Third, spectral averaging and exhaustion yield (1.4) (Theorem 3.2); Corollary 3.3 gives (1.6) and the exponentially decaying eigenbasis. □

1.1 The quantitative input and its source

Let $T_N = (\mathbb{Z}/3^N\mathbb{Z})^2$ carry the periodic nearest-neighbor operator with the original iid uniform potentials. Its distance d_N is the shortest-coordinate maximum distance.

Hypothesis 1.2 (Quantitative fixed-energy input)**.** *For every $E \in (-4-h, 4+h)$ there exist $A_E, a_E > 0$ and $N_E \in \mathbb{N}$ such that*

$$\mathbb{E}|G_{T_N}(x,y;E)|^q \leq A_E e^{-a_E d_N(x,y)}, \qquad q = 1/128, \quad N \geq N_E, \quad x,y \in T_N. \tag{1.7}$$

The constants may depend on h, E, but not on N, x, y.

This is the statement claimed in [1, Theorem 10.1, source label `boot:fractional`]: taking $s = 1/64$ in its exponent $s/2$ gives $q = 1/128$. The estimate concerns the physical torus operator, after auxiliary probes have been removed, at each fixed $E \in (-4-h, 4+h)$. Its constants and volume threshold may depend on h, E. The source constructs calibrated resonant cells, controls their transmission and closure, and organizes them into a screened hierarchy; its final resolvent estimate is the interface used here [1, Sections 4–10].

We assume this quantitative statement. The proof below establishes its implication for dynamical localization and does not independently verify the source's multiscale construction. The repository's stated Lean scope covers bounded self-adjointness and the spectral interval, and excludes the pure-point conclusion [2]. The chosen scales are finite and existential; no numerical localization length is extracted.

Pure-point spectral type alone gives $Q_{\Sigma_h}(x,y) = \sum_{\lambda\in\mathrm{spec}_{\mathrm{pp}}(H_h)} |\langle \delta_x, P_{\{\lambda\}}(H_h)\delta_y\rangle|$, with no exponential bound on this sum or its expectation. The additional spatial control comes from (1.7).

The Anderson model originates in [3]. Two standard rigorous approaches are multiscale analysis [4, 5] and the fractional-moment method [6, 7]. The latter is the route used in the transfer theorem. Spectral averaging and resolvent criteria for pure-point spectrum are developed in [8]; fractional-moment consequences for correlations and dynamics are discussed in [9, 11]. In two dimensions, the Bernoulli result of [12] concerns energies near a spectral edge. The

independent, possibly nonstationary setting of [13] likewise treats a bottom-of-spectrum regime. Such statements illustrate why a theorem covering the entire weak-disorder spectrum needs its precise quantitative input. These references supply methods and established implications; they are not cited as proofs of Hypothesis 1.2 at arbitrary weak disorder throughout the two-dimensional spectrum.

1.2 Proof plan and conventions

Section 2 proves the resolvent bound; Section 3 turns it into localization. After Hypothesis 1.2, the main input is the fractional-moment finite-volume criterion of [7], stated below in the needed form. Spectral averaging and the spectral readout are cited in Section 3, with their hypotheses checked. The local calculations use Schur complements (block Gaussian elimination), Cauchy–Schwarz, one-variable integration and compactness:

$$\begin{aligned} (1.7) &\Longrightarrow B_{\Lambda_L}(E) \le \tfrac{1}{4} \quad \text{for each interior } E, \quad L = L(E), \\ \text{continuity, endpoints and compactness} &\Longrightarrow (2.13), \\ \text{spectral averaging and exhaustion} &\Longrightarrow (1.4) \Longrightarrow (1.6). \end{aligned} \tag{1.8}$$

For vectors, $|x|_1 = |x_1| + |x_2|$ and $|x|_\infty = \max(|x_1|, |x_2|)$; $\|\cdot\|$ is the operator norm. Constants may depend on the fixed disorder h. At each prescribed real E, finite physical inverses exist almost surely, since

$$\det(H_{h,\Omega} - E) = \prod_{x\in\Omega} v_x + \text{terms of smaller total degree}, \qquad \mathbb{P}_h\{\det(H_{h,\Omega} - E) = 0\} = 0. \tag{1.9}$$

This is an energywise null-set statement. Integrability of fractional powers will justify the real-energy averages below. For background on the spectral theorem and localization methods, see [14, 11].

2 From torus decay to a uniform resolvent bound

Assume the pointwise torus estimate (1.7). Write

$$T_N = (\mathbb{Z}/3^N\mathbb{Z})^2, \qquad d_{T_N}(x,y) = \max_{j=1,2} \min_{k\in\mathbb{Z}} |x_j - y_j + 3^N k|.$$

Suppose that I is nonempty and compact, and $0 < q < 1/2$ is fixed. For each $E \in I$ suppose there are $A_E, a_E > 0$ and $N_E \in \mathbb{N}$ such that

$$\mathbb{E}|G_{T_N}(x,y;E)|^q \le A_E e^{-a_E d_{T_N}(x,y)}, \qquad N \ge N_E. \tag{2.1}$$

No uniformity of A_E, a_E, N_E in E is assumed. Real finite-volume resolvents are defined outside the null event on which the tested energy is an eigenvalue. Set $p = q/2 < 1/4$; this range lets us use the published decoupling criterion directly and includes $q = 1/128$.

2.1 Averaging and the finite-volume criterion

Two elementary inequalities explain the exponents: pole integrability for $0 < r < 1$ [6], and Cauchy–Schwarz for $q = 2p$:

$$\begin{aligned} \int_{-\varepsilon}^{\varepsilon} |t|^{-r}\, dt &= \frac{2\varepsilon^{1-r}}{1-r} < \infty, \\ \mathbb{E}|G_1 G_2|^p &\le \left(\mathbb{E}|G_1|^q\, \mathbb{E}|G_2|^q\right)^{1/2}. \end{aligned}$$

Lemma 2.1 (Uniform-law averaging and decoupling)**.** *For any finite real symmetric hopping matrix with independent diagonal entries uniform on $[-h, h]$, every resolvent entry satisfies*

$$\sup_{z\in\mathbb{C}\setminus\mathbb{R}} \mathbb{E}|G(x,y;z)|^r \le K_r, \qquad K_r = \frac{4h^{-r}}{1-r}, \quad 0<r<1. \tag{2.2}$$

The same bound holds at every prescribed real energy. For $0 < p < 1/4$ the law also satisfies the ASFH decoupling condition with a finite constant $D_p = D_p(h)$. We may use

$$\kappa_p = K_p D_p^2 < \infty \tag{2.3}$$

in the finite-volume criterion below.

Verification of the cited hypotheses. The uniform law has bounded support and $\mathbb{P}(|v-a| < \varepsilon) \le \varepsilon/h$. Apply [7, Lemma B.1, (B.2), (B.5)–(B.8)] with $\tau = 1$, $\lambda = 1$ and concentration constant $\kappa_1 = 1/h$:

$$\mathbb{E}|G(x,y;z)|^r \le \frac{(4/h)^r}{1-r} \le K_r.$$

Bounded support and [7, Lemma C.3] give the finite decoupling constant for $p < 1/4$. The convention $\widetilde{C}_p = C_p D_p^2$ in [7, after (1.8)], with $C_p \le K_p$, gives (2.3). □

The finite-volume test sums propagation from the center to crossing bonds. A strict margin gives a contraction at each iteration. The boundary distance below becomes ordinary lattice distance for fixed sites in exhausting boxes.

Theorem 2.2 (ASFH finite-volume criterion, specialized)**.** *For the iid uniform Anderson model, let $0 < p < 1/4$ and let $\Lambda \subset \mathbb{Z}^2$ be finite with $0 \in \Lambda$. Define*

$$\Gamma(\Lambda) = \{(u,u') : u \in \Lambda,\ u' \notin \Lambda,\ |u-u'|_1 = 1\}.$$

If, at a prescribed real or complex z,

$$B_\Lambda(z) := (1+\kappa_p|\Gamma(\Lambda)|)^2 \sum_{(u,u')\in\Gamma(\Lambda)} \mathbb{E}|G_\Lambda(0,u;z)|^p < 1, \tag{2.4}$$

then fractional moments decay exponentially in the bulk, uniformly over finite regions Ω. Their finite-region distance is

$$d_\Omega(x,y) = \min\{|x-y|_1, \operatorname{dist}(x,\partial\Omega) + \operatorname{dist}(y,\partial\Omega)\}. \tag{2.5}$$

One may use the inner boundary $\partial\Omega = \{u \in \Omega : \exists u' \notin \Omega, |u-u'|_1 = 1\}$; this only weakens the estimate if the exterior-boundary convention is used. Distances to an empty boundary are $+\infty$. In particular, for $\Lambda = [-L, L]^2$ and $B_\Lambda(z) \le 1/2$, one may take

$$\mathbb{E}|G_\Omega(x,y;z)|^p \le 4K_p \exp\left\{-\frac{\log 2}{2L+2} d_\Omega(x,y)\right\}. \tag{2.6}$$

This is Theorem 1.2 and its iteration in equations (2.25)–(2.32) of [7], with the upper constants in Lemma 2.1.

For a test bounded by $b = 1/2$, the cited iteration has prefactor $K_p b^{-2} = 4K_p$ and step radius at most $2L+1$; weakening the rate to $(\log 2)/(2L+2)$ gives (2.6). For fixed x, y and boxes $\Omega_n \uparrow \mathbb{Z}^2$, $d_{\Omega_n}(x,y) = |x-y|_1$ eventually.

2.2 A local test from the torus estimate

Proposition 2.3 (Torus input supplies a Dirichlet test)**.** *For each prescribed $E_0 \in I$,* (2.1) *supplies a finite L with $B_{\Lambda_L}(E_0) \le 1/4$, where $\Lambda_L = [-L, L]^2$. There is $\delta > 0$ for which $B_{\Lambda_L}(z) \le 1/2$ whenever $|z - E_0| \le \delta$. Both L and δ may depend on E_0.*

Proof. First fix $E = E_0$ and abbreviate $A = A_{E_0}$, $a = a_{E_0}$, $N_0 = N_{E_0}$. For $N \ge N_0$, embed Λ_L in T_N with $L = \lfloor 3^N/4 \rfloor$ and $g_L = |\Gamma(\Lambda_L)| = 8L + 4$. Cutting its crossing bonds gives

$$G_{\Lambda_L}(0, u; E) = G_{T_N}(0, u; E) + \sum_{(v,v') \in \Gamma(\Lambda_L)} G_{T_N}(0, v'; E) G_{\Lambda_L}(v, u; E).$$

For $u \in \partial\Lambda_L$ and $(v, v') \in \Gamma(\Lambda_L)$, $d_{T_N}(0, u), d_{T_N}(0, v') \ge L$. Subadditivity and Cauchy–Schwarz, with $q = 2p$, yield

$$\begin{aligned} \mathbb{E}|G_{\Lambda_L}(0, u; E)|^p &\le \big(\mathbb{E}|G_{T_N}(0, u; E)|^q\big)^{1/2} \\ &\quad + \sum_{(v,v') \in \Gamma(\Lambda_L)} \big(\mathbb{E}|G_{T_N}(0, v'; E)|^q \, \mathbb{E}|G_{\Lambda_L}(v, u; E)|^q\big)^{1/2} \\ &\le \sqrt{A}(1 + g_L\sqrt{K_q}) e^{-aL/2}. \end{aligned} \tag{2.7}$$

The resolvents need not be independent. Since $g_L \le 12L$ for $L \ge 1$,

$$B_{\Lambda_L}(E) \le C(h, E) L^4 e^{-aL/2} \longrightarrow 0. \tag{2.8}$$

Since $L^4 e^{-aL/2} \to 0$ and $\lfloor 3^N/4 \rfloor \to \infty$, choose one $N \ge N_0$ for which (2.8) is at most $1/4$.

For this fixed box, the resolvent identity gives

$$G_{\Lambda_L}(0, u; z) - G_{\Lambda_L}(0, u; E) = (z - E) \sum_{w \in \Lambda_L} G_{\Lambda_L}(0, w; z) G_{\Lambda_L}(w, u; E).$$

Lemma 2.1 and Cauchy–Schwarz imply

$$\mathbb{E}|G_{\Lambda_L}(0, u; z) - G_{\Lambda_L}(0, u; E)|^p \le |z - E|^p (2L + 1)^2 K_q. \tag{2.9}$$

Use $\big||w|^p - |w'|^p\big| \le |w - w'|^p$ and choose

$$\delta = \big[4(1 + \kappa_p g_L)^2 g_L (2L + 1)^2 K_q\big]^{-1/p}. \tag{2.10}$$

Equations (2.9)–(2.10) increase the test by at most $1/4$ when $|z - E| \le \delta$. Hence $B_{\Lambda_L}(z) \le 1/2$, and Theorem 2.2 applies with $\mu_0 = (\log 2)/(2L + 2)$. □

2.3 The endpoints and one finite energy cover

Lemma 2.4 (The two spectral endpoints)**.** *Put $\Sigma_h = [-4 - h, 4 + h]$ and $E_\pm = \pm(4 + h)$. For every finite Dirichlet Ω and $0 < p < 1$,*

$$\mathbb{E}|G_\Omega(x, y; E_\pm)|^p \le \frac{(2h)^{-p}}{1 - p} \, r_h^{p|x-y|_1}, \qquad r_h = \frac{2}{h} \log(1 + h/2) < 1. \tag{2.11}$$

Proof. First work at E_+ with $W_x = h - v_x \sim \mathrm{Unif}[0, 2h]$, $D_{xx} = 4 + W_x$, $P = D^{-1} A_\Omega$, and $\alpha_x = 4/(4 + W_x)$. The nonnegative Dirichlet Laplacian and Neumann expansion give

$$\begin{aligned} K := E_+ - H_{h,\Omega} &= 4\mathrm{I} - A_\Omega + \mathrm{diag}(W), \\ K &\ge \mathrm{diag}(W) > 0, \qquad (K^{-1})_{yy} \le W_y^{-1}, \\ K^{-1} &= \sum_{n \ge 0} P^n D^{-1}, \qquad \|P\|_\infty \le \max_{x \in \Omega} \alpha_x < 1. \end{aligned}$$

Let $\mathbb{E}_x^{\mathrm{rw}}$ denote the ordinary nearest-neighbor walk, τ_y its first hit of y, and τ_{Ω^c} its exit time. First-hit decomposition gives

$$H_\omega(x,y) = \mathbb{E}_x^{\mathrm{rw}}\left[\mathbf{1}_{\{\tau_y<\tau_{\Omega^c}\}}\prod_{k=0}^{\tau_y-1}\alpha_{X_k}\right],$$
$$(K^{-1})_{xy} = H_\omega(x,y)(K^{-1})_{yy}, \qquad 0 \le H_\omega \le 1.$$

The product excludes y, so $H_\omega(x,y)$ is independent of W_y. For a path stopped at its first hit, let n_z be its departure counts. Independence and $0 \le \alpha_z \le 1$ give

$$\#\{z : n_z \ge 1\} \ge |x-y|_1,$$
$$\mathbb{E}\prod_z \alpha_z^{n_z} \le \prod_{z:n_z\ge 1}\mathbb{E}\alpha_z = r_h^{\#\{z:n_z\ge1\}},$$
$$\mathbb{E}\alpha_z = \frac{1}{2h}\int_0^{2h}\frac{4}{4+w}\,dw = \frac{2}{h}\log(1+h/2) = r_h < 1.$$

Tonelli and the ordinary first-hit path probabilities yield

$$\mathbb{E}H_\omega(x,y) \le \mathbb{E}_x^{\mathrm{rw}}\left[\mathbf{1}_{\{\tau_y<\tau_{\Omega^c}\}} r_h^{\#\{z:n_z\ge1\}}\right] \le r_h^{|x-y|_1}.$$

Independence and Jensen now give

$$\begin{aligned}\mathbb{E}|(K^{-1})_{xy}|^p &\le \mathbb{E}W_y^{-p}\,\mathbb{E}H_\omega(x,y)^p \\ &\le \frac{(2h)^{-p}}{1-p}(\mathbb{E}H_\omega(x,y))^p \\ &\le \frac{(2h)^{-p}}{1-p}r_h^{p|x-y|_1}.\end{aligned}$$

For $x = y$, take $H_\omega(y,y) = 1$. At E_- use

$$(Uu)(x) = (-1)^{x_1+x_2}u(x), \qquad -UH(v)U = H(-v),$$
$$|G_{H(v)}(x,y;E_-)| = |G_{H(-v)}(x,y;E_+)|.$$

and $W_x = h + v_x$. This proves (2.11). □

Proposition 2.5 (One finite energy cover)**.** *Let $I \subset \mathbb{R}$ be nonempty and compact. Suppose each $E \in I$ admits a finite L_E with $B_{\Lambda_{L_E}}(E) \le 1/4$. Then there are finitely many $E_j \in I$, radii $\delta_j > 0$ and scales L_j such that*

$$I \subset \bigcup_{j=1}^{J}(E_j - \delta_j/2, E_j + \delta_j/2), \qquad L_{\max} = \max_j L_j < \infty. \tag{2.12}$$

On an open complex neighborhood of I, uniformly over finite Ω,

$$\mathbb{E}|G_\Omega(x,y;z)|^p \le 4K_p e^{-\mu_0 d_\Omega(x,y)}, \qquad \mu_0 = \frac{\log 2}{2L_{\max}+2}. \tag{2.13}$$

The pointwise torus input supplies the premise on any compact set of input energies. If that input holds at every interior energy, the premise also holds for $I = \Sigma_h$. In particular, this applies to Hypothesis 1.2 with $q = 1/128$.

Proof. First suppose the torus input holds at every interior energy. Proposition 2.3 supplies the interior tests. At the endpoints, Lemma 2.4 gives, with $g_L = 8L + 4$,

$$B_{\Lambda_L}(E_\pm) \le (1+\kappa_p g_L)^2 g_L \frac{(2h)^{-p}}{1-p} r_h^{pL} \longrightarrow 0.$$

Thus both endpoint tests are at most $1/4$ at sufficiently large scales.

For the general assertion, (2.9)–(2.10) give a disk $D_E = \{z : |z - E| < \delta_E\}$ on which $B_{\Lambda_{L_E}}(z) \le 1/2$. Select a finite subcover by the intervals of half these radii. For $z \in D_{E_j}$, Theorem 2.2 gives

$$\mathbb{E}|G_\Omega(x,y;z)|^p \le 4K_p e^{-(\log 2)d_\Omega(x,y)/(2L_j+2)} \le 4K_p e^{-\mu_0 d_\Omega(x,y)}.$$

The finite union $\mathcal{U} = \bigcup_j D_{E_j}$ contains the compact set I, so $\delta = \frac{1}{2}\operatorname{dist}(I, \mathbb{C}\setminus\mathcal{U}) > 0$ and $I_\delta := \{z : \operatorname{dist}(z, I) < \delta\} \subset \mathcal{U}$. This proves (2.13) with one energy cover. □

3 From resolvent decay to dynamical localization

3.1 Spectral averaging

Use the cross spectral measure $\mu_{xy}(B) = \langle \delta_x, P_B(H)\delta_y\rangle$. The spectral theorem [14, Chapter 3] and [11] give

$$G(x,y;z) = \int \frac{\mu_{xy}(dE)}{E - z},$$

$$\langle \delta_x, f(H)P_J(H)\delta_y\rangle = \int_J f(E)\,\mu_{xy}(dE),$$

$$Q_J(x,y) = \sup_{|f|\le 1}\left|\int_J f\,d\mu_{xy}\right| = |\mu_{xy}|(J).$$

In finite volume this is $\sum_{\lambda\in J \text{ distinct}} |\langle \delta_x, P_{\{\lambda\}}\delta_y\rangle|$, so eigenvalue simplicity is unnecessary.

Lemma 3.1 (Bounded-uniform spectral interpolation)**.** *For every finite Dirichlet region Ω, every Borel set J of finite Lebesgue measure, and $0 < p < 1$,*

$$\mathbb{E}Q_{J,\Omega}(x,y) \le \left[(2h)^{p-1}\int_J \mathbb{E}|G_\Omega(x,y;E)|^p\,dE\right]^{1/(2-p)}. \tag{3.1}$$

Here $Q_{J,\Omega}$ is the supremum over bounded Borel multipliers of modulus at most one for $H_{h,\Omega}$.

Specialization of the published spectral-averaging estimate. Apply [7, Appendix A, (A.9)–(A.10), (A.12)] with $\alpha = p$. Independent single-site resampling preserves the iid law, its displacement is at most $2h$, and the conditional density is bounded by $(2h)^{-1}$. Thus

$$\big(\mathbb{E}Q_{J,\Omega}(x,y)\big)^{2-p} \le (2h)^p(2h)^{-1}\int_J \mathbb{E}|G_\Omega(x,y;E)|^p\,dE,$$

which is (3.1). The cyclic-projection normalization (A.9) requires no eigenvalue simplicity. □

3.2 Passing to the lattice

Theorem 3.2 (Dynamical localization from pointwise fractional moments)**.** *Assume (2.1), or take $I = \Sigma_h$ and assume that input at every interior energy. Let $L = L_{\max}$ be selected by Proposition 2.5, and set*

$$\mu = \frac{\log 2}{(2-p)(2L+2)}, \qquad C = \max\left\{1, [4K_p(2h)^{p-1}|I|]^{1/(2-p)}\right\}. \tag{3.2}$$

Then

$$\mathbb{E}Q_I(x,y) \le Ce^{-\mu|x-y|_1}, \qquad x,y\in\mathbb{Z}^2. \tag{3.3}$$

In particular, for $I = [-4,4]$ the prefactor may be taken as

$$C = \max\left\{1, \left[\frac{128\,2^{p-1}}{1-p}h^{-1}\right]^{1/(2-p)}\right\}.$$

Proof. Put $D = [4K_p(2h)^{p-1}]^{1/(2-p)}$. Equations (2.13) and (3.1) give, for boxes $\Omega_n \uparrow \mathbb{Z}^2$ and Borel $J \subset I_\delta \cap \mathbb{R}$,

$$\mathbb{E}Q_{J,\Omega_n}(x,y) \le D|J|^{1/(2-p)} e^{-\mu d_{\Omega_n}(x,y)}.$$

For fixed x, y, $d_{\Omega_n}(x,y) = |x-y|_1$ eventually.

The cross spectral measures converge weakly along this exhaustion; see [7, proof of Theorem A.1], or the common-core and functional-calculus results [14, Lemma 6.36, Theorem 6.31]. Here the zero extensions $H_{h,\Omega_n} \oplus 0$ are uniformly bounded and agree eventually with H_h on finitely supported vectors. For open $J \subset I_\delta \cap \mathbb{R}$, lower semicontinuity of total variation and Fatou therefore give

$$\mathbb{E}|\mu_{xy}|(J) \le \liminf_n \mathbb{E}|\mu_{xy,\Omega_n}|(J) \le D|J|^{1/(2-p)} e^{-\mu|x-y|_1}.$$

Take $J_k = \{E : \operatorname{dist}(E, I) < \min(\delta/2, 1/k)\}$. Then $J_k \downarrow I$ and $|J_k| \downarrow |I|$. Since $|\mu_{xy}|(\mathbb{R}) \le 1$, continuity from above and dominated convergence give

$$\mathbb{E}Q_I(x,y) \le D|I|^{1/(2-p)} e^{-\mu|x-y|_1} \le Ce^{-\mu|x-y|_1}.$$

We use the finite-volume estimate from the proof of [7, Theorem A.1]. The theorem itself assumes ordinary-distance decay in every finite region; the exhaustion above transfers our boundary-distance estimate to the ordinary lattice distance. □

3.3 Time evolution and eigenvectors

Corollary 3.3 (Dynamical moments and spectral localization)**.** *Under the same hypothesis, for every $r > 0$ and $y \in \mathbb{Z}^2$,*

$$\mathbb{E}\sup_{t\in\mathbb{R}} \sum_{x\in\mathbb{Z}^2} (1+|x-y|_1)^r \left|\langle \delta_x, e^{-itH_h} P_I(H_h)\delta_y\rangle\right|^2 \le C \sum_{x\in\mathbb{Z}^2} (1+|x-y|_1)^r e^{-\mu|x-y|_1} < \infty. \quad (3.4)$$

The spectral subspace $P_I(H_h)\ell^2(\mathbb{Z}^2)$ is almost surely pure point and admits an orthonormal basis of exponentially decaying eigenvectors.

Proof. Since $|\langle \delta_x, e^{-itH_h} P_I(H_h)\delta_y\rangle|^2 \le Q_I(x,y)$, Tonelli and (3.3) give

$$\mathbb{E}\sup_t \sum_x (1+|x-y|_1)^r |\langle \delta_x, e^{-itH_h} P_I(H_h)\delta_y\rangle|^2 \le C \sum_x (1+|x-y|_1)^r e^{-\mu|x-y|_1} < \infty.$$

The spectral implication of a summable infinite-volume correlator bound is the argument in [10, Appendix B(ii)–(iii)], applied here with one particle to (3.3). It gives pure-point spectrum and a complete family of exponentially decaying projected coordinate vectors; the pure-point step uses RAGE [14, Theorem 5.8], see also [15]. Within each eigenspace, Gram–Schmidt on that countable spanning family uses finite linear combinations and preserves exponential decay. This yields an orthonormal eigenbasis without a simplicity assumption. □

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